

Experiment 7: Random Sampling and Sampling Distribution

Objective:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

Steps:

1. **Generate a Population:**
 - Create a population of data with a specified distribution (e.g., normal distribution).
2. **Random Sampling:**
 - Perform random sampling from the population to create multiple samples of different sizes.
 - Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. **Sampling Distribution:**
 - Plot histograms or density plots of sample statistics (e.g., sample means).
 - Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. **Central Limit Theorem (Optional):**
 - Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

Example Code:

python

Copy code

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Step 1: Generate a population (e.g., normal distribution)
```

```
population_mean = 50
```

```
population_std = 10
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```
population_size = 100000
```

```
population = np.random.normal(population_mean, population_std, population_size)
```

```
# Step 2: Random sampling
```

```
sample_sizes = [30, 50, 100] # different sample sizes to consider
```

```
num_samples = 1000 # number of samples for each sample size
```

```

sample_means = {}

for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i+1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='Population Mean')
    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()

```

OUTPUT:

